

Python Basics

1. Introduction

Python is a high-level, interpreted programming language known for its simplicity and readability. It supports multiple programming paradigms such as object-oriented, functional, and procedural.

2. Features

- Easy to learn and use
- Open source and cross-platform
- Rich standard library
- Interpreted and dynamically typed
- Supports integration with C, C++, and Java

3. Data Types

- int, float, complex
- str (string)
- list, tuple, set, dict
- bool (True/False)

4. Control Statements

```
```python
if x > 0:
 print("Positive")
elif x == 0:
 print("Zero")
else:
 print("Negative")
```
```

5. Functions

```
```python
def greet(name):
 return f"Hello, {name}!"
```
```

6. Object-Oriented Programming

```
```python
class Car:
 def __init__(self, brand, model):
 self.brand = brand
 self.model = model
 def show(self):
 print(self.brand, self.model)
```
```